Project Report

1. Introduction

Build a machine learning model that predicts outcomes using historical data and evaluate its performance.

Example Projects

- House Price Prediction (Linear Regression)
- Student Performance Prediction
- Customer Churn Prediction
- Disease Prediction
- Loan Approval Prediction

Example:

Predict House Prices

Input Features:

- Area
- Bedrooms
- Bathrooms
- Location Score

Output:

- House Price

2. Dataset Description

Source and features

Sources:

- Kaggle
- UCI Machine Learning Repository
- Government Open Data Portals

Example Dataset:

Area	Bedrooms	Bathrooms	Price
1200	2	2	250000
1500	3	2	320000
1800	3	3	400000

3. Data Preprocessing — Cleaning and Preparation
List steps like handling missing values, encoding categories, scaling numbers, and splitting data. Example: "Imputed missing TotalCharges with median, one-hot encoded categorical variables, standardized numerical features, 80/20 train-test split."

4. Model Building — Algorithms Used
Name the models and briefly justify each. Example: "Logistic Regression (baseline), Random Forest (robust), XGBoost (high performance)."

5. Evaluation — Metrics and Results
Pick metrics relevant to your problem, then show results in a table. Example: "Used accuracy, precision, recall, F1-score, ROC AUC. The table shows XGBoost leading in F1-score (0.65) and ROC AUC (0.87)."

6. Visualization — Charts and Graphs
List what you plotted and why. Example: "Confusion matrix for error analysis, ROC curve for threshold trade-off, feature importance plot to identify top predictors like Contract and MonthlyCharges."

7. Conclusion — Best-Performing Model
State the winner and why. Example: "XGBoost performed best overall based on F1-score and ROC AUC. Month-to-month contracts and high monthly charges were top churn predictors. Future work could address class imbalance or test retention strategies."